# Naming Chemical Compounds

**SET : nonmetal-nonmetal compounds**

Rules:
1. if hydrogen is present, it is named first
2. if no hydrogen, the solid nonmetal is named first
3. if only two elements, name of second is shortened and suffix '-ide' is added to the shortened form
4. if only two elements, prefixes 'mon-', 'di-', 'tri-', 'tetra-' etc. are used on the name of the second to indicate how many atoms of it there are in the formula.
5. if more than two elements, first is named and this is followed by the name of the radical.

examples
$HCl$ = hydrogen chloride
$CO_2$ = carbon dioxide
$H_2CO_3$ = hydrogen carbonate

$PCl_3$
$P_2O_3$
$CO$
$CCl_4$
$NBr_5$
$NO_2$
$N_2O_3$
$PBr_3$
$SO_3$
$SCl_6$
$HBr$
$PCl_5$
$N_2O_5$
$CO_2$
$SF_6$

**SET : metal-nonmetal compound; two elements only**

Rules:
1. the metallic element is named first
2. the name of the nonmetal is shortened
3. the suffix '-ide' is added to this shortened name

examples
$NaCl$ = sodium chloride
$MgBr_2$ = magnesium bromide

$BaCl_2$
$AlCl_3$
$Li_2O$
$FeS$
$PbO$
$MgF_2$
$CaS$
$CuF_2$
$MnBr_4$
$Fe_2O_3$
$ZnI_2$
$Pb_3O_4$
$LiBr$
$AgBr$
$NiCl_2$

- many chemical compounds contain a chemical radical
- a chemical radical is a group of atoms which tend to stay together when the compound reacts with another compound
- the names of many chemical compounds include the name of the radical they contain
- the names of the most common radicals are shown in the table:

| Name of radical | Formula of radical as it appears in chemical compounds |
|---|---|
| ammonium | $NH_4$ |
| carbonate | $CO_3$ |
| hydrogen carbonate | $HCO_3$ |
| hydrogen sulfate | $HSO_4$ |
| hydrogen sulfite | $HSO_3$ |
| hydroxide | $OH$ |
| nitrate | $NO_3$ |
| sulfate | $SO_4$ |
| sulfite | $SO_3$ |

**SET : metal-nonmetal compound; three or more elements**

Rules: 1. the metallic element is named first
2. the chemical radical is named second

examples $FeSO_4$ = iron sulfate
$Mg(NO_3)_2$ = magnesium nitrate

$K_2CO_3$
$Zn(NO_3)_2$
$CaCO_3$
$Zn(HSO_4)_2$
$Cu(OH)_2$
$BaSO_4$
$FeCO_3$
$KHSO_4$
$CuSO_3$
$Al_2(SO_3)_3$
$Mg(OH)_2$
$CaSO_4$

**SET 4: Common names**

There are no rules to follow here. You just have to know what the common names are.

| | |
|---|---|
| Carbon monoxide | $CO$ |
| Carbon dioxide | $CO_2$ |
| Nitrogen dioxide | $NO_2$ |
| Sulfur dioxide | $SO_2$ |
| Hydrogen sulfide | $H_2S$ |
| Ammonia | $NH_3$ |
| Methane | $CH_4$ |
| Hydrochloric acid | $HCl$ |
| Nitric acid | $HNO_3$ |
| Sulfuric acid | $H_2SO_4$ |
| Carbonic acid | $H_2CO_3$ |
| Ethanoic (acetic) acid | $CH_3COOH$ |
| Phosphoric acid | $H_3PO_4$ |
| Salt | $NaCl$ |
| Caustic soda | $NaOH$ |
| Sodium bicarbonate | $NaHCO_3$ |
| Lime | $CaO$ |
| Marble chips | $CaCO_3$ |
| Lime water | $Ca(OH)_2$ |
| Copper sulfate | $CuSO_4$ |
| Water | $H_2O$ |